

Single-Image 3D Generation

A Comparative Evaluation of Three Feed-Forward Methods

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Why single-image 3D matters

Where 3D content is needed today

E-commerce

AR / VR

Robotics

Game pipelines

Time to one asset

Manual artist	4–8 hours
Feed-forward model	< 10 sec

▷ **Question:** which feed-forward model should you actually use?

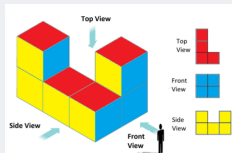
The task: 1 image $\rightarrow$ 3D mesh

RGB image
 $1 \times 512 \times 512$

?

3D mesh
+ texture

Depth ambiguity



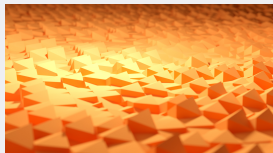
Many shapes project to the same image

The back is unseen



Half the surface is never observed

Texture vs. geometry



Shadows can be mistaken for shape

3D primer: five terms used in this talk

Mesh

Vertices + triangle faces.

Stores geometry; texture stored per-vertex or via UV map.

NeRF MLP

A network mapping a 3-D point

$(x, y, z) \rightarrow (\text{density}, \text{colour})$.

Mesh is extracted from the density field afterwards.

Triplane

Three orthogonal 2-D feature grids (XY, YZ, XZ).

Compactly encodes a 3-D scene; cheap lookup.

Rendering

Project the 3-D representation
from a chosen camera $\rightarrow$ 2-D image.

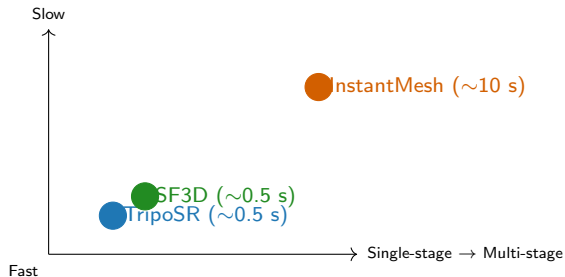
Multi-view consistency

Render the same object from N angles.

If renders disagree (Janus heads, blank backs),
the 3-D representation is incoherent.

► That's everything you need. No 3D background assumed.

Three methods, three priors



	TripoSR	InstantMesh	SF3D
Year	2024	2024	2025
Paradigm	LRM	Diff + LRM	LRM + MatN
Output	vertex colour	vertex colour	UV + PBR
Speed	~ 0.5 s	~ 10 s	~ 0.5 s

LRM = large reconstruction model (a transformer that maps an image to a 3-D representation in one pass).

PBR = physically-based rendering — the asset has separate albedo / roughness / metallic maps so it can be *relit*.

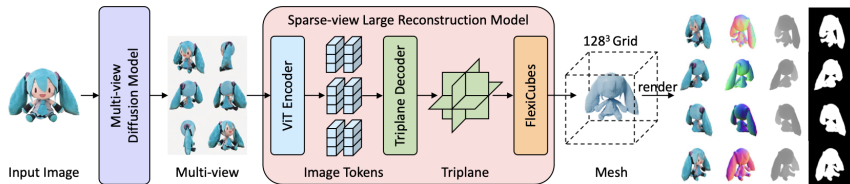
TripoSR — single-stage triplane → NeRF



- DINOv1 ViT → image tokens
- Cross-attention → 64×64 triplane
- NeRF MLP reads density + colour
- Marching cubes → vertex-coloured mesh

► 64^2 triplane is coarse → aliases on thin parts. No material model.

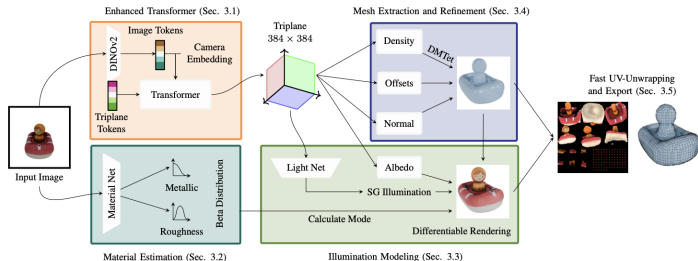
InstantMesh — hallucinate views, then reconstruct



- Stage 1: Zero123++ diffusion → **6 novel views**
- Stage 2: sparse-view LRM → mesh
- FlexiCubes iso-surface + depth / normal supervision
- End-to-end ~ 10 s; only watertight method here

► Diffusion can disagree with itself → noise propagates into geometry. $\sim 20\times$ slower.

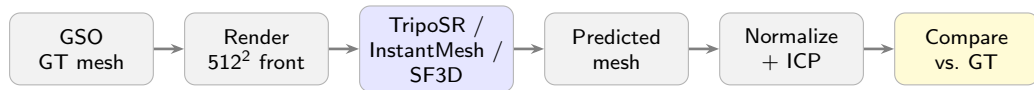
SF3D — geometry + material + lighting



- Higher-resolution triplane (vs. TripoSR)
- **MaterialNet**: albedo / roughness / metallic
- **LightNet**: estimates illumination
- Vertex offsets smooth the iso-surface
- UV unwrap → PBR-ready GLB

✓ Only method whose output is directly drop-in into a renderer / game engine. CVPR'25.

How we evaluate (one input, three pipelines, same metrics)



Dataset

100 objects from Google Scanned Objects.
Categories: household, apparel, electronics, toys, packaging.

Pipeline

Render canonical front view → feed all three models.
Both meshes normalized to unit bbox → 24-rotation search → ICP.

Metrics — image & alignment

Metric	Dir.	Definition
PSNR	↑	$10 \log_{10}(1/\text{MSE})$ between front render and input — pixel match.
SSIM	↑	Structural similarity in local windows; robust to small misalignments.
LPIPS	↓	Distance in AlexNet feature space — best correlates with human texture judgement.
CLIP-sim	↑	Cosine of CLIP embeddings (input vs. render) — <i>semantic</i> identity match.
Sil-IoU	↑	Silhouette IoU: $ \hat{S} \cap S_{\text{in}} / \hat{S} \cup S_{\text{in}} $. Catches <i>scale</i> / <i>pose</i> mismatches that pixel metrics miss.
Multi-view CLIP	↑	Mean CLIP similarity across $N=8$ rendered views. Low $\Rightarrow$ Janus heads, blank backs.

Metrics — geometry vs. ground truth

Metric	Dir.	Definition (after both meshes are normalized to unit bbox)
Chamfer-L1	↓	$\bar{d}_{p \rightarrow r} + \bar{d}_{r \rightarrow p}$, i.e. mean nearest-neighbour distance both ways between sampled point clouds. Sensitive to outliers.
F-Score @ τ	↑	Sample point matched if it is within τ of the other surface; report F1 of precision/recall. We use $\tau \in \{1\%, 2\%, 5\%\}$ of the bbox diagonal.
NC (normal consistency)	↑	$\frac{1}{2} (\mathbb{E} \cos \theta_{p \rightarrow r} + \mathbb{E} \cos \theta_{r \rightarrow p})$ where θ is the angle between matched face normals. <i>Surfaces face the same way</i> , not just sit close.
W % (watertight)	↑	Fraction of meshes that are closed manifolds. Required for volume / 3-D printing / fluid sim.

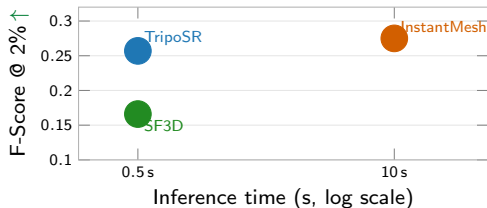
▷ We always report both Chamfer *and* F-Score: a few stray vertices wreck Chamfer; F-Score is robust to that.

Results — geometry vs. GT ($n=100$)

Method	CD ↓	F@1% ↑	F@2% ↑	NC ↑	W% ↑
TripoSR	0.129	0.134	0.257	0.642	0 %
InstantMesh	0.132	0.152	0.275	0.627	38 %
SF3D	0.185	0.084	0.166	0.526	0 %

CD =

Chamfer-L1 (unit-bbox); NC = normal consistency; W% = watertight fraction.



F@2 %: InstantMesh 0.275

~ Multi-view diffusion sees 6 hallucinated views $\Rightarrow$ recovers occluded surfaces.

Chamfer / NC: TripoSR wins

~ Single coherent NeRF density $\Rightarrow$ no view-aggregation artefacts.

W %: InstantMesh 38 %

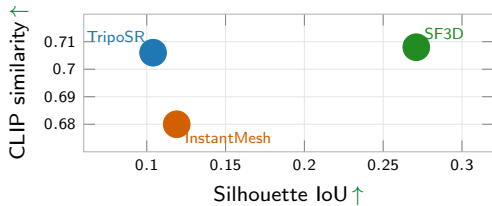
~ FlexiCubes outputs closed manifolds; the other two extract via marching cubes and leak holes.

All scores low, F@1 %: 0.08–0.15

~ Single-image 3D is still hard at strict thresholds.

Results — appearance & alignment ($n=100$)

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	CLIP $\uparrow$	Sil-IoU $\uparrow$	M
TripoSR	11.19	0.756	0.455	0.706	0.104	0.917
InstantMesh	11.15	0.727	0.507	0.680	0.119	0.917
SF3D	11.31	0.731	0.491	0.708	0.271	0.917



Sil-IoU: SF3D 0.27 ($2.5\times$ others)

~ UV-PBR pipeline preserves the input scale; others routinely produce mesh that is wrong size or pose.

LPIPS / SSIM: TripoSR

~ Coherent NeRF $\Rightarrow$ smoother textures and structures pixel-by-pixel.

MV CLIP: SF3D 0.917

~ Most coherent from all sides.

CLIP-sim front: tied SF3D / TripoSR

~ Object identity preserved across all three.

Property-based slice — 30 hand-labelled objects

F-Score @ 2 % ($\uparrow$), by surface type

Surface	n	TripoSR	InstantMesh	SF3D
smooth	11	0.329	0.273	0.129
textured	8	0.279	0.411	0.146
detailed	11	0.244	0.227	0.168

	Material	n	TripoSR	InstantMesh	SF3D
LPIPS ($\downarrow$), by material	matte	16	0.378	0.439	0.413
	glossy	11	0.428	0.481	0.474
	mixed	3	0.558	0.616	0.585

Textured surface $\rightarrow$ InstantMesh

$\rightsquigarrow$ 0.411 vs. 0.279 / 0.146 — printed-graphics packaging hits the multi-view diffusion sweet spot.

Smooth / detailed $\rightarrow$ TripoSR

$\rightsquigarrow$ Coherent NeRF generalises to “generic” shapes.

Mixed (metallic, $n=3$) $\rightarrow$ all fail

$\rightsquigarrow$ LPIPS $\sim$ 0.6 across the board; specular cues confuse every method.

Subset: 30/100 objects hand-labelled in data/property_labels.csv. Full breakdown across thickness / surface / material in runs/eval/by_property.md.

Qualitative — three objects, three stories

Object	Input	TripoSr	InstantMesh	SF3D
Elephant (organic)				
Dragon (complex)				
Toaster (metallic)				

Predictable failure modes

TripoSR



Thin structures

64^2 triplane smooths fork tines / wire away.

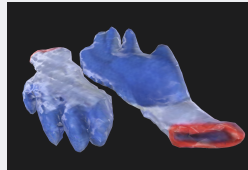
InstantMesh



Reflective surfaces

Diffusion hallucinates inconsistent novel views.

SF3D



Spatially-varying material

MaterialNet predicts one global (ρ, m) .

▷ Failures are *architectural*, not random. They line up with the metrics on the previous slides.

When to pick which method

TripoSR

Default. Smooth
or detailed objects.
Best perceptual texture (LPIPS).

InstantMesh

Textured packaging,
complex occlusion.
Need watertight
output (10 s OK).

SF3D

Output goes into a
renderer / engine.
Need correct pose & relight.

Open problems (all three)

- Transparent / reflective objects
- Wiry / thin geometry
- Spatially-varying materials
- Strict tolerance ($\tau < 1\%$): F-Score ~ 0.1

- **No method dominates.** Each wins a different metric family.
- **Architecture predicts behaviour.** NeRF $\Rightarrow$ smooth; diffusion $\Rightarrow$ filled-in occlusions; PBR $\Rightarrow$ better pose.
- **Property type matters more than averages.** Pick the model from the object class.
- **Single-image 3-D is still hard at strict tolerance** ($F@1\% \leq 0.15$).

Thank you.

Code, data labels, and reports: see the GitHub repository.